Modeling Memories, Predicting Prospections: Automated Scoring of Autobiographical Detail Narration using Large Language Models

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Abstract

The Autobiographical Interview is a widely used tool for examining memory and related cognitive functions. It provides a standardized framework to differentiate between internal details, representing the episodic features of specific events, and external details, including semantic knowledge and other non-episodic information. This study introduces an automated scoring model for autobiographical memory and future thinking tasks, using large language models (LLMs) that can analyze personal event narratives without preprocessing. Building on the traditional Autobiographical Interview protocol, we fine-tuned a LLaMA-3 model to identify internal and external details at a narrative level. The model was trained and tested on narratives from 284 participants across three studies, spanning past and future thinking tasks, multiple age groups, and collected in lab and virtual interviews. Results demonstrate strong correlations with human scores of up to $r = 0.87$ on internal and up to $r = 0.84$ on external details, indicating the model aligns as closely with human raters as they do with each other. Additionally, as evidence of the algorithm's construct validity, the model replicated known age-related trends that cognitively normal older adults generate fewer internal and more external details than younger adults across three datasets, finding this age group difference even in one dataset where human raters did not. This automated approach offers a scalable alternative to manual scoring, making large-scale studies of human autobiographical memory more feasible.

Keywords Autobiographical memory; Future event thinking; Automated scoring; Natural language processing; Large language model

Introduction

The Autobiographical Interview (Levine et al., 2002) is a widely used method for examining the content of autobiographical event memories, which are memories of unique moments in time and place. In this structured procedure, participants are prompted to recall specific personal events in detail, and their narratives are scored according to a technical manual that specifies rules for identifying and categorizing details as either internal or external. Internal details pertain to episodic information specific to the event, such as time, place, emotions, and perceptual experiences. External details, on the other hand, include semantic information, repetitions, comments, or details about events other than the central one being recalled. This separation is valuable because it allows researchers to quantify the specificity of episodic memory retrieval and assess the balance between episodic and semantic content in personal narratives (Levine, 2004; Levine et al., 2002).

The Autobiographical Interview has been extensively employed across various research domains, particularly in the study of aging and cognitive decline. Levine et al. (2002) first demonstrated that cognitively normal older adults produce fewer internal details and more external details than young adults when recalling autobiographical memories, indicating an age-related reduction in episodic memory specificity. This foundational finding has been replicated and confirmed in numerous subsequent studies. For example, Simpson et al. (2023) conducted a meta-analysis, which reinforced that cognitively normal older adults consistently show decreased internal detail recall and increased external details relative to young adults.

Extending this research to populations at risk for Alzheimer's disease (AD), Grilli et al. (2018) found that cognitively normal middle-aged and older adults carrying the apolipoprotein E ε4 (APOE ε4) allele—a genetic risk factor for AD—produced fewer internal details than non-carriers, suggesting that reduced episodic specificity may serve as an early marker of cognitive decline before clinical symptoms manifest. Acevedo-Molina et al. (2023) replicated

and extended these results by showing that APOE ε4 carriers generate fewer internal details compared to non-carriers while remembering and imagining future events. Related work has shown similar outcomes using cutoffs on neuropsychological screening measures (Li-Chay-Chung et al., 2024). Additionally, Simpson et al. (2023) demonstrated through their meta-analysis that clinical populations diagnosed with both mild cognitive impairment (MCI) and dementia exhibit pronounced decreases in internal details, underscoring the sensitivity of the Autobiographical Interview in capturing memory impairments linked to neurodegenerative conditions.

Beyond aging and risk for cognitive decline, the Autobiographical Interview has been adapted to diverse contexts in neurological and psychological research, such as for assessing memory deficits in individuals with brain injuries (Esopenko & Levine, 2017; Race et al., 2011) and depression (Söderlund et al., 2014). Importantly, as alluded to by the research on AD risk, the Autobiographical Interview is not limited to assessing memory of past events but has also been adapted for other tasks, such as future thinking. For instance, Addis et al. (2008) employed it to study how individuals imagine future events, finding that age-related reductions in internal details during past event recall are similarly present in future event simulations.

The Autobiographical Interview's widespread use in over 300 studies (Simpson et al., 2023) has provided an array of valuable insights. However, the manual scoring process remains time-consuming and labor-intensive. Scoring typically takes around ten minutes per memory, with each participant often contributing multiple memories (Levine et al., 2002; van Genugten & Schacter, 2024). This requirement significantly constrains large-scale research, as the resources required for manual scoring limit the number of participants that can be feasibly included (van Genugten & Schacter, 2024). Consequently, most studies using the Autobiographical Interview have relatively small sample sizes, averaging about 30–50 participants (Simpson et al., 2023).

Previous automation approaches

Given the significant time and resource demands of manually scoring the Autobiographical Interview, along with the reduced generalizability of findings based on their small sample sizes, researchers have pursued various methods to automate and streamline the process. Initial efforts centered on computational models to estimate internal and external detail counts, while recent approaches have taken advantage of advances in natural language processing and machine learning.

Peters et al. (2017) made an early attempt to automate scoring by analyzing linguistic features—such as word counts, emotion word frequency, and parts of speech—to build regression models that could predict the number of internal (episodic) and external (semantic) details in participants' narratives. While innovative, these models were limited in their accuracy, often capturing overall narrative length rather than effectively distinguishing between internal and external content. Around the same time, Takano et al. (2017, 2018, 2019) worked on automating the scoring of the Autobiographical Memory Test (AMT), a simpler task involving the classification of memories as either specific or general (Williams & Broadbent, 1986). By training classifiers on word frequencies and parts of speech, they achieved high accuracy in this simpler classification task, demonstrating the potential of natural language processing in memory scoring. However, their methods did not address the complexities of the Autobiographical Interview, where detailed segmentation and classification of both internal and external details are required.

More recently, Mistica et al. (2024) developed a machine learning model based on the BERT language model (Devlin et al., 2019) to automatically classify autobiographical memories into broader categories: specific, categoric, extended, semantic associate, and omission. By training on over 17,000 human-scored memories from diverse demographic groups, they aimed to generalize across populations and reduce scoring variability. While their work advanced the

automation of memory assessment, it did not specifically adhere to the detailed scoring guidelines of the Autobiographical Interview established by Levine et al. (2002), focusing instead on broader memory classifications.

Building on advances in natural language processing, van Genugten & Schacter (2024) introduced an automated scoring procedure specifically for the Autobiographical Interview. They fine-tuned the DistilBERT language model (Sanh et al., 2019), a compact variant of BERT (Devlin et al., 2019), to classify sentences based on the proportion of internal content, categorizing them into classes of 0%, 50%, 75%, or 100% internal content. After classifying each sentence, they calculated the internal and external word counts by multiplying the total words in a sentence by the predicted percentage of internal content. For instance, a ten-word sentence predicted to be 50% internal contributes five words to the internal word count and five to the external. Summing these across all sentences in a narrative provided total internal and external word counts. They found that these word counts closely correlated with manually identified internal and external details, effectively using word counts as proxies for detail counts and simplifying the scoring process. Despite this advancement, the method relies on clean, well-punctuated text, requiring careful transcription and preprocessing. Uninformative speech elements, such as filler words, need to be removed to reduce noise. Additionally, applying word count at the sentence level may not capture the fine-grained detail segmentation prescribed by the original scoring manual and does not consider the contextual meaning of words relative to each other within sentences and across the narrative.

While previous automation attempts have contributed valuable insights and tools, challenges remain in fully automating the scoring of the Autobiographical Interview, particularly for spoken language. Existing methods either require substantial manual effort or do not fully align with the detailed scoring criteria established by Levine et al. (2002). The approach developed by van Genugten & Schacter (2024) represents the most advanced attempt

to date, but further refinements are needed to address its limitations and enhance its utility in large-scale research.

Present study

Recent advancements in large language models (LLMs) have created new possibilities for automating complex language tasks that traditionally required extensive human effort. Leveraging these developments, the present study introduces a novel automated scoring model for the Autobiographical Interview. Our model processes entire (spoken or written) narratives without any preprocessing, such as cleaning transcripts or removing filler words and interviewer prompts. By capturing the relationships between words and sentences within the narratives, it focuses on meaningful content, potentially offering better generalization across diverse datasets.

The motivation for this new model stems from the limitations of previous automation attempts, which either necessitated substantial manual preprocessing or did not fully align with the detailed scoring criteria essential for cognitive research. Our approach addresses these issues by utilizing an efficient computational architecture that allows for dynamic adaptation to different scoring tasks by using specialized model adapters. This flexibility not only enhances the model's applicability but also facilitates expansion for future research needs.

The primary objective of this study was to assess whether our automated scoring model can accurately identify internal and external details in autobiographical memory tasks using raw, unprocessed transcripts, achieving a level of alignment with human scorers comparable to that observed between two human scorers. Demonstrating this alignment is crucial for the model's potential to replace manual scoring in cognitive research. Additionally, we aimed to show that the model can replicate typical research findings related to age differences in narrative detail generation. Specifically, we expect the model to empirically recover the well-established pattern where cognitively normal older adults produce fewer internal details and more external

details compared to young adults, in line with the differences identified by human scorers (Levine et al., 2002; Simpson et al., 2023). By replicating these age-related findings as an example, we evaluate an element of the automated scoring approach's construct validity. Furthermore, we sought to determine whether the model can generalize to non-memory tasks, such as future thinking. These are areas where the Autobiographical Interview has been effectively adapted in previous research (Addis et al., 2008), and we thereby aimed to assess its versatility across different types of narrative analyses.

Methods

Data

We use three datasets in this study for training and evaluation purposes. Dataset A (Hernandez et al., 2024) and Dataset B (Acevedo-Molina et al., 2023) each had 20 subjects randomly held out to maintain separate sets for training and evaluation, representing approximately 20% of the total sample in Dataset A and 25% in Dataset B. Dataset C (unpublished) was exclusively used for evaluation, with no overlap in training, ensuring fully independent testing.

All narratives were scored using the Autobiographical Interview manual (Levine et al., 2002), categorizing details as either "internal" (episodic, including time, place, perceptual, thought, and emotion elements) or "external" (including semantic, repetitions, metacognitive statements, and external events). Two trained raters independently scored each narrative, and their scores were averaged to ensure reliable categorization.

Dataset A: Videoconference Autobiographical Interview (Hernandez et al., 2024)

Hernandez et al. (2024) examined memory recall from life events among young adults (ages 18–34) and cognitively normal older adults (ages 61–83) using a virtual, videoconference Autobiographical Interview. The study included 103 participants, with 54 young and 49 older

adults, predominately being White non-Hispanic according to self-reported race and ethnicity. Neuropsychological screening ensured older adults performed within age-appropriate norms. Cognitive eligibility among older adults was determined through a comprehensive neuropsychological assessment, using a profile-based approach (Bondi et al., 2014).

Following the Autobiographical Interview manual (Levine et al., 2002), participants recalled five personal memories, including events from early childhood, adolescence, early adulthood, and the past year, with older adults additionally recalling a memory from middle adulthood. Each memory was to represent an event lasting less than 24 hours, with initial recall followed by a general prompt for elaboration. Narratives were recorded, transcribed, and independently analyzed by two trained raters following the guidelines of the Autobiographical Interview scoring manual (Levine et al., 2002).

Dataset B: Autobiographical past and future event thinking (Acevedo-Molina et al., 2023), scene memory (unpublished), and scene construction (unpublished)

Acevedo-Molina et al. (2023) examined autobiographical memory and future thinking in a sample of 82 cognitively normal older adults (ages 60–80), predominantly being White non-Hispanic, including 39 APOE ε4 carriers and 43 non-carriers. Similar to (Hernandez et al., 2024), eligibility was determined through a comprehensive neuropsychological assessment, using a profile-based approach (Bondi et al., 2014) to ensure participants met criteria for normal cognitive functioning.

The autobiographical memory and future thinking task, adapted from Addis et al. (2010), involved two sessions. In the first session, participants generated 16 vivid memories from the past five years, each containing a distinct person, object, and place. Five memories were then preserved for recall trials, while details from the remaining memories were recombined to create unique triplets for future event trials. In the second session, participants either revisited selected memories or imagined new events incorporating these recombined details, providing

detailed narratives over a four-minute period for each trial. This process resulted in a total of five past and five future thinking trials.

Additionally, participants completed a scene construction task, imagining themselves in detailed scenarios—such as a meadow or a busy train platform—and describing sensory experiences, including sights, sounds, smells, and touch. They spent four minutes elaborating on each scene, creating vivid mental experiences without referencing past or future events. Finally, participants completed a scene memory task, which involved viewing 360-degree videos of various global scenes and later recalling each scene's visual details after short and long delays to assess memory retention over time. Data from the scene construction and scene memory tests were not previously published by our group. Narratives were recorded, transcribed, and independently analyzed by two trained raters following the guidelines of the Autobiographical Interview scoring manual (Levine et al., 2002).

Dataset C: Past and future autobiographical event thinking and the Autobiographical Interview

In this ongoing study, 99 participants completed a past and future autobiographical event thinking tasks and the Autobiographical Interview, along with neuropsychological tests to assess cognitive functioning within age-appropriate norms. The sample included in the present study comprised 18 young adults (ages 20–33) and 81 cognitively normal older adults (ages 60–84), mainly being White non-Hispanic.

In the past and future autobiographical event thinking task, participants were given a cue word and instructed to either recall a specific event from the past few years (past cues) or imagine a specific event that could realistically happen to them in the next few years (future cues). Each participant completed five past and five future trials. The Autobiographical Interview followed traditional procedures, including time period cues, prompting participants to retrieve and describe five personal memories within a five-minute recall period per memory.

Narratives were recorded, transcribed, and independently analyzed by two trained raters following the guidelines of the Autobiographical Interview scoring manual (Levine et al., 2002).

Model training

We fine-tuned a LLaMA-3 large language model using a continuous regression approach to assess narrative scoring for internal and external details. Leveraging quantization and LoRA, we optimized computational efficiency and adapted the model for this scoring task (explained in more detail below). No data preprocessing was applied to the training and evaluation narratives. Prior to training, test sets were held out for independent evaluation after the training process was completed. Separate adapters for internal and external scoring were trained using the same hyperparameter settings and training process.

Large Language Models

Since the first conceptual introduction by Vaswani et al. (2017), large language models (LLMs) have been successfully applied in a variety of natural language processing (NLP) tasks (Kaddour et al., 2023). LLMs are based on the transformer architecture, which was designed to address the limitations of recurrent neural networks (RNNs) and convolutional neural networks (CNNs) in handling sequential data and long-range dependencies (Karita et al., 2019; Vaswani et al., 2017). The architecture's key innovation, the self-attention mechanism, allows the model to weigh the significance of different words in a sentence relative to each other, capturing contextual relationships more effectively than previous architectures (Shaw et al., 2018).

The attention mechanism works by assigning attention scores to words based on their relevance to each other. For instance, in the sentence, "The cat sat on the mat because it was soft", the word "it" refers to "the mat", not "the cat". The attention mechanism enables the

model to focus on "the mat" when interpreting "it", thereby capturing the semantic relationship between these words. This capability extends to understanding longer contexts, allowing transformers to maintain contextual awareness across multiple sentences (Shaw et al., 2018; Vaswani et al., 2017).

Different model families have emerged from the transformer architecture, each tailored for specific applications. OpenAI's GPT series (Radford et al., 2018), beginning with GPT-1 and evolving through GPT-4 (OpenAI, 2023), has been widely adopted in applications ranging from text summarization to code generation (Dale, 2021). In research applications, various BERT-based models such as BERT (Devlin et al., 2019), DistilBERT (Sanh et al., 2019), RoBERTa (Liu et al., 2019), and DeBERTa (He et al., 2021) have been used extensively.

Recently, the LLaMA (Large Language Model Meta AI) series has gained attention for its efficiency and high performance on a range of language tasks (Touvron et al., 2023). LLaMA-3, the latest iteration, builds upon its predecessors by incorporating advanced training techniques and larger, more diverse datasets, thereby enhancing its ability to generate and process coherent and contextually relevant text (Dubey et al., 2024). A benefit of LLaMA is its open-source nature, in contrast to the full GPT models, which are limited to paid users. Moreover, LLaMA can be run locally on a desktop computer, eliminating reliance on commercial cloud platforms and providing notable advantages in privacy, cost, and accessibility. This model exemplifies the ongoing trend of developing more efficient and capable transformer-based architectures equipped to handle increasingly complex tasks in natural language processing.

Low-rank adaptation fine-tuning

Fine-tuning is a powerful technique for adapting LLMs to perform specialized tasks by training on task-specific datasets. By updating a subset or all of the model's parameters, fine-tuning can enable models to capture nuances within a target domain, leading to high performance on tasks

like text classification, summarization, or specific types of narrative analysis. For single-task fine-tuning, it is often possible to achieve notable improvements with as few as 50-100 additional training examples, provided the pre-trained model is already aligned with the target task (Chung et al., 2022; Howard & Ruder, 2018).

However, full fine-tuning of LLMs presents challenges due to the extensive computational resources and time required to adjust all model parameters. This process can be especially impractical for the largest models, where updating billions of parameters becomes a logistical hurdle. To address these limitations, researchers have developed parameter-efficient fine-tuning techniques that selectively modify only a small portion of the model parameters while leaving the majority of the model weights unchanged. These have been shown to achieve similar task-specific performance improvements as full fine-tuning, but with considerably reduced computational overhead (Houlsby et al., 2019).

In particular, low-rank adaptation (LoRA) has gained popularity as a parameter-efficient solution. This approach works by adding low-rank matrices to the pre-trained model during fine-tuning, essentially creating a series of smaller, task-specific adjustments. Instead of updating the entire weight matrix, LoRA inserts a low-rank decomposition that focuses on the parameters most relevant to the target task. By reducing the number of trainable parameters, LoRA enables fine-tuning of large models on limited hardware. This method allows researchers to achieve high accuracy on specialized tasks without the resource demands of full model fine-tuning (Hu et al., 2021).

To further reduce computational overhead, quantization can be applied alongside LoRA to compress model weights. Quantization lowers the precision of model weights by converting 16- or 32-bit floating-point values to more compact formats, such as 8-bit integers. This reduces memory usage and speeds up computations, while preserving model accuracy to a high degree.

This approach, known as qLoRA, allows large models to operate efficiently on limited hardware while achieving strong task performance (Dettmers et al., 2023).

An important benefit of LoRA and qLoRA is their modularity. Since the base model remains unchanged during fine-tuning, each trained adapter matrix is relatively small, enabling multiple adapters to be created for different tasks. During inference, the base model can be loaded into memory once, and only the adapters are swapped to effectively yield different models as needed. This flexibility allows for efficient task switching and is particularly valuable in scenarios requiring multiple specialized models. In our case, separate adapters could be trained for identifying internal and external details, which were then applied sequentially to analyze narratives in a computationally efficient way (explained in more detail below).

Training parameters

We used the Hugging Face Transformers library to fine-tune our model (Wolf et al., 2020) and we performed the training on a single NVIDIA RTX A6000 GPU with 48 GB of memory, supported by a system with 45 GB of RAM and 8 CPU cores. For our base model, we selected the LLaMA-3 model with 8 billion parameters, due to its strong performance across standard benchmarks and suitability for text scoring tasks (Dubey et al., 2024).

Many NLP text scoring applications are handled as discrete classification tasks, such as sentiment analysis, topic classification, and spam detection (Minaee et al., 2021). However, the scoring of internal and external details in our study spans a wide range of values, which would produce numerous, sparsely populated classes and a highly imbalanced training set if modeled as a classification problem. For this reason, we opted to frame our scoring task as a continuous regression problem. Although less common, this method has been successfully applied in previous research on similar datasets (Sun & Wang, 2024; Yang et al., 2020). To adapt the LLaMA-3 model for regression in the Transformers class, we employed a regression head that outputs continuous values instead of discrete classes (Wolf et al., 2020).

We explored different hyperparameter configurations but found that varying the main parameters within reasonable ranges had marginal impact. As a result, we settled on a configuration that balanced simplicity and performance. Specifically, we set the learning rate to 1e-5 and used a batch size of 64, with an automatic batch-sizing feature to dynamically adjust the batch size based on memory limits as needed. The low learning rate allowed for steady, gradual optimization, reducing the risk of overshooting the optimal solution (Goodfellow et al., 2016; Wolf et al., 2020). All other hyperparameters were left at their default values, ensuring consistency and reducing the complexity. To enhance computational efficiency, we implemented quantization using the bitsandbytes package from the Hugging Face library, reducing the model's precision to 8-bit. Additionally, we applied LoRA from the Hugging Face PEFT library, using a rank of 128. This value determines the rank of each matrix that LoRA inserts alongside the model's main weights, and it is high enough to capture essential task-specific patterns while remaining compact enough to avoid overfitting (Hu et al., 2021). The alpha parameter, set to 256, serves as a scaling factor that adjusts the impact of the low-rank updates on the model's overall structure (Hu et al., 2021). To maintain consistency and ensure comparability, we trained two separate adapters—one for internal details and one for external details—using identical hyperparameters across both tasks.

Data preparation and training process

For training and evaluation, no data preprocessing was applied to the narratives. The narratives were used in their raw transcript form, including all parts spoken by the interviewer, such as questions, probes, and clarifying responses to participants' questions. Additionally, the transcripts retained timestamps, placeholders for sensitive details, and filler words. First, the raw text was tokenized, a process that converts words and punctuation into numerical representations, making it readable by machine learning algorithms. Then, the tokenized data was fed directly into the model. A shortened fictional example for a narrative is shown below:

Q: [1:15] Now can you tell me about me about an event at a specific time and place from your teenage years?

[1:20] Yeah, sure, uh, I remember this one time when my friends and I decided to camp out at [Local Park Name]. We, uh, packed all our stuff, like tents and food, and rode our bikes there after school. I think there were five of us? It was me, [Female Name 1], [Male Name 2], and a couple of others. Anyway, we set up right next to this big tree, and we were just, uh, hanging out, telling stories… even roasted marshmallows on this little fire pit we made. It got pretty late, probably around midnight, and then we realized we forgot sleeping bags. We just ended up using our jackets and, you know, talking until we fell asleep. [2:05]

General Probe: [2:10] Any other details you remember from that event?

[2:12] Oh, yeah, it was freezing! [2:13]

For the test sets, narratives from 20 subjects were randomly held out from both Dataset A (Hernandez et al., 2024) and Dataset B (Acevedo-Molina et al., 2023) prior to training. For Dataset B, we selected narratives from the memory and future autobiographical event thinking task as test sets, allowing us to validate our hypothesis that the automated scoring model would generalize across tasks, including future-oriented thinking. The scene construction and scene memory tasks were included only in the training set to increase sample size and introduce diverse narrative types, potentially enhancing the model's generalizability. Dataset C was held out entirely, as data collection was still ongoing during training, resulting in a test set with narratives from 99 subjects.

After removing the held-out test sets, we split the remaining data, allocating 90% for model training and 10% as a validation set. During training, the model ran inference on the validation set every 100 steps. Key metrics—training loss, validation loss, root mean squared error (RMSE), and R-squared—were monitored to assess performance. Training was halted once

these metrics plateaued, indicating no further improvement on the validation set. This process was first completed for the internal details adapter, and then repeated for the external details adapter using the same training and evaluation settings.

Model evaluation and statistical analyses

We employed multiple statistical methods to evaluate the automated scoring model's performance, comparing it to human raters and assessing its ability to capture established age-related patterns in narrative detail generation across multiple datasets. Through these comparisons, we aimed to validate the model's robustness and its potential for application in cognitive research.

Comparison to manual scorers

To evaluate the model's performance, we applied automated scoring to the holdout sets from Dataset A (Hernandez et al., 2024) and Dataset B (Acevedo-Molina et al., 2023), and the complete Dataset C. Each narrative was scored for internal and external details by the model, and these scores were compared against scores produced by manual scorers.

To assess how well the model aligns with human raters, we calculated Pearson correlations between the model's scores and human ratings. We compared the model's scores with both the average score of the two human raters and each rater's individual score for each dataset. We used the correlation between the two human raters as a benchmark. If the model's correlation with each rater is close to the correlation between the raters themselves, we consider it to perform comparably to human scoring. To further clarify this alignment, we included 95% confidence intervals (CI) for the correlations and created scatter plots to visually depict the relationship between the model's and raters' scores, highlighting any alignment or differences across datasets.

For Dataset C, we analyzed factors that might explain differences between the automated scores and the average manual scores. Specifically, we conducted linear regression and correlation analyses to determine whether discrepancies were associated with participant variables (e.g., age and education level), narrative variables (e.g., total word count), or the degree of disagreement between manual raters, quantified as the absolute difference between their scores. This analysis provided insight into the conditions under which the model's scores might deviate from human judgments.

Additionally, to assess whether the model selectively attends to relevant content, we compared scores generated from unprocessed transcripts with those from preprocessed versions. Preprocessing included the automated removal of interviewer speech and filler words, the replacement of placeholders for sensitive information, and punctuation correction. This allowed us to evaluate any impact that preprocessing may have on the model's scoring of narrative details.

All analyses were conducted in Python 3.12. Pearson correlations were calculated using the SciPy library, while scatter plot visualizations were generated using Seaborn and Matplotlib.

Age group differences

As evidence of the construct validity and practical usefulness of the automated scoring model, we evaluated its ability to replicate established research findings, specifically age-related differences in detail generation patterns. In line with previous studies (Levine et al., 2002; Simpson et al., 2023), young adults are expected to generate more internal details, while older adults are anticipated to produce more external details.

We conducted a mixed-design analysis of variance (ANOVA) for each manual scorer, the average of the manual scorers, and the automated scoring. This ANOVA examined main effects for the between-subjects factor of age group (young vs. older adults) and the within-subjects factor of detail type (internal vs. external), as well as their interaction. Significant interactions

were further explored using independent samples *t*-tests to compare internal and external scores between age groups for each scorer. Given the large sample sizes for all comparisons, independent samples *t*-tests are appropriate despite Shapiro-Wilk tests indicating that most samples are not normally distributed. To further validate our findings, we also conducted non-parametric Mann-Whitney *U* tests, which confirmed all conclusions drawn from the independent samples *t*-tests. Effect sizes are reported as Cohen's *d* for the independent samples *t*-tests.

In addition to these inferential tests, we calculated descriptive statistics—including mean, median, and standard deviation (*SD*)—and assessed normality and variance assumptions using Shapiro-Wilk and Levene's tests.

All analyses were performed in Python 3.12. The mixed ANOVA was conducted using the Pingouin library, while descriptive statistics and tests were calculated in SciPy. Visualizations for these analyses, specifically box plots highlighting age group differences, were generated in Matplotlib and Seaborn.

Results

Narratives from training holdout sets

We first evaluated the model's performance on holdout sets from Dataset A (Hernandez et al., 2024) and Dataset B (Acevedo-Molina et al., 2023). As noted, in each dataset, subjects were randomly selected for the holdout sets, and all narratives from them were excluded before training to ensure the model had no prior exposure to these data.

Dataset A: Videoconference Autobiographical Interview (Hernandez et al., 2024)

For the Autobiographical Interview narratives in Dataset A, the model demonstrated high alignment with human ratings in scoring internal details. The correlation between human raters

for internal details, serving as the benchmark, was strong at r = .76 (95% CI [.66, .83]), indicating consistent scoring between them (Fig. 1a). When the model's scores were compared to the average human score, the correlation increased to r = .87 (95% CI [.82, .91]) (Fig. 1b), suggesting that the model's scoring aligns more closely with the combined human ratings than the human raters do with each other. Correlations between the model and each individual human rater were similarly robust, with r = .85 (95% CI [.78, .90]) for the first rater (Fig. 1c) and r = .80 (95% CI [.71, .86]) for the second rater (Fig. 1d).

For external details in Dataset A, human raters again showed high consistency, with a benchmark correlation of r = .86 (95% CI [.80, .91]) (Fig. 1e). The model's alignment with the average human score for external details was slightly lower, achieving r = .78 (95% CI [.69, .85]) (Fig. 1f), but still indicating a strong level of correlation. When compared with individual raters, the model maintained similarly strong correlations, achieving r = .75 (95% CI [.66, .83]) for the first rater (Fig. 1g) and r = .75 (95% CI [.65, .82]) for the second rater (Fig. 1h). All correlations were significant at $p < .001$. Overall, the model achieves strong correlation levels comparable to the consistency observed between two human raters, highlighting its reliability as an alternative scoring method.

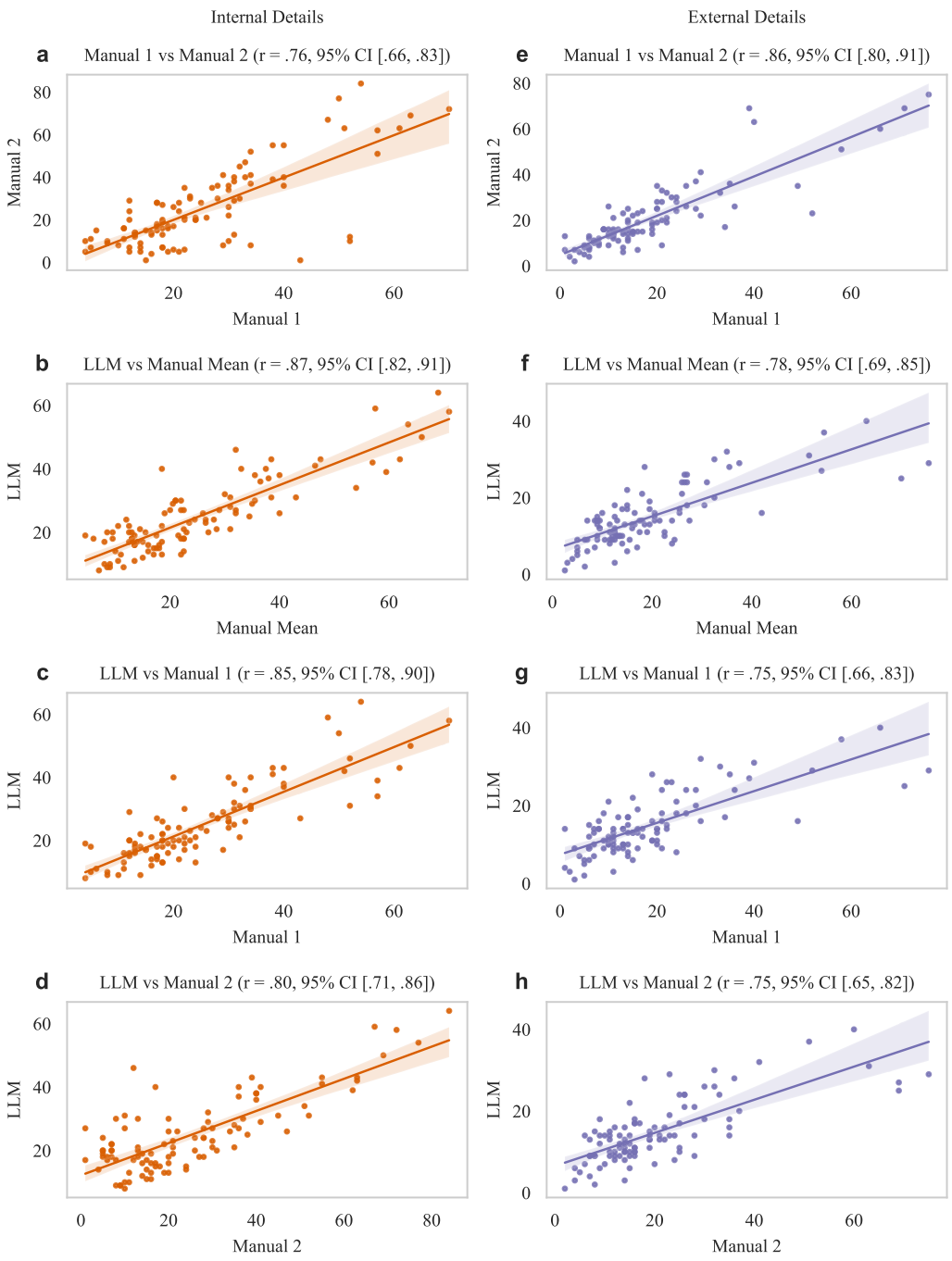

Figure 1. Model alignment with human raters on internal and external detail scoring in Dataset A (Hernandez et al., 2024). For internal details, the model's alignment with the average human score (b), the first individual scorer (c), and the second individual scorer (d) is comparable to the alignment between the two human scorers (a). For external details, the model's alignment with the average human score (f) and with each individual scorer (g-h) also shows strong alignment, similar to the correlation observed between human raters (e). All correlations are significant at $p < .001$.

Dataset B: Autobiographical event memory (Acevedo-Molina et al., 2023)

Similar to the results for Dataset A, the model displayed strong correlations with human raters across both internal and external detail types for the memory task in Dataset B. The model's scores for internal details demonstrated a correlation of r = .83 (95% CI [.76, .89]) with the manual mean, r = .82 (95% CI [.74, .88]) with the first scorer, and r = .76 (95% CI [.66, .83]) with the second scorer, while the correlation between the two human raters was r = .80 (95% CI [.71, .86]) (Fig. 2a). For external details, the model's scores correlated at r = .77 (95% CI [.68, .84]) with the manual mean, r = .70 (95% CI [.58, .79]) with the first scorer, and r = .75 (95% CI [.65, .83]) with the second scorer, while the human raters achieved a correlation of r = .76 (95% CI [.67, .84]) (Fig. 2b). All correlations were significant at $p < .001$. Overall, these high correlations confirm that the model reliably replicates human scoring in memory-focused tasks, achieving a level of alignment comparable to that observed between two manual scorers. A detailed visualization of the correlations using scatter plots, similar to those presented for Dataset A in Fig. 1, is provided in Supplementary Fig. 1.

Dataset B: Autobiographical future event thinking (Acevedo-Molina et al., 2023)

In the future thinking task from Dataset B, the model again showed strong performance. Correlations for internal details reached r = .83 (95% CI [.76, .89]) with the manual mean, r = .78 (95% CI [.69, .85]) with the first scorer, and r = .79 (95% CI [.70, .85]) with the second scorer, while the two human raters had a correlation of r = .78 (95% CI [.68, .85]) (Fig. 2a). For external details, the model's correlations were r = .84 (95% CI [.77, .89]) with the manual mean, r = .82 (95% CI [.74, .88]) with the first scorer, and r = .74 (95% CI [.63, .82]) with the second scorer, compared to a correlation of r = .73 (95% CI [.62, .81]) between the two human raters (Fig. 2b). All correlations were significant at $p < .001$. The model’s strong performance on the future thinking task suggests it can generalize beyond memory tasks to future-oriented thinking, as discussed by Addis et al. (2008). This versatility highlights the model’s potential

for broader applications across various narrative-based assessments in cognitive research. A detailed visualization of the correlations using scatter plots, similar to those presented for Dataset A in Fig. 1, is provided in Supplementary Fig. 2.

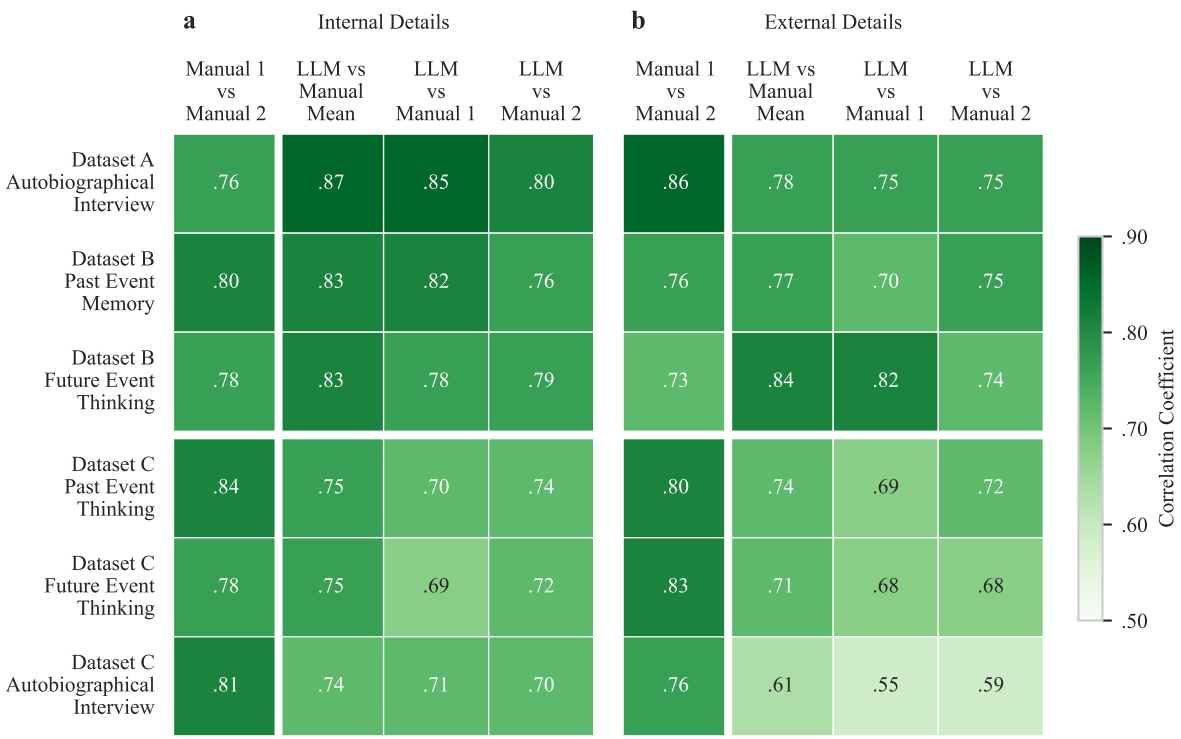

Figure 2. Correlation of model and human ratings on internal and external detail scoring across multiple datasets. Panel (a) presents the correlation coefficients for internal details, while panel (b) shows the results for external details. All correlations are significant at $p < .001$. Each panel compares the consistency between human raters (Manual 1 vs. Manual 2) and the alignment of the model (LLM) with human scores, including comparisons with the average human score and with each individual rater. The model demonstrates strong alignment with human raters across datasets, particularly for internal details, with correlations generally high in Dataset A and Dataset B. Consistent with Dataset C being a fully new dataset—which, although procedurally similar, likely differs in key aspects such as the sample, context, and timing of data collection—the model's performance is somewhat lower. This is particularly evident for external details in the Autobiographical Interview, where correlations drop to moderate levels.

Narratives from training-independent study

To assess the model's generalizability, we evaluated its performance on narratives from a fully independent dataset, Dataset C. This evaluation tested the model's ability to score both past- and future-oriented autobiographical event thinking using data from studies it had not previously encountered during training. While generalization to new datasets is a critical goal, it is important to note that a non-trivial loss in prediction accuracy is typically expected when moving beyond the training data due to potential differences in samples, context, or time. As with the other datasets, we compared the model's scores for internal and external details to human ratings, using the correlation between two human raters as a performance benchmark.

Dataset C: Past autobiographical event thinking

For internal details in the past autobiographical event thinking task, the correlation between human raters was strong, $r = .84$ (95% CI [.81, .86]), setting a high benchmark for comparison (Fig. 3a). The model's correlation with the human mean was somewhat lower, at $r = .75$ (95% CI [.71, .79]) (Fig. 3b), but still indicates strong alignment. Correlations with individual raters were $r = .70$ (95% CI [.65, .74]) with the first rater (Fig. 3c) and $r = .74$ (95% CI [.69, .77]) with the second rater (Fig. 3d), further demonstrating robust alignment with human assessments. While performance on internal details slightly declined compared to the training holdout data, the model remains well-aligned with human scoring, even on this new dataset.

For external details, human raters achieved a benchmark correlation of $r = .80$ (95% CI [.77, .83]) (Fig. 3e). The model's alignment with the human mean for external details was slightly lower, at $r = .74$ (95% CI [.70, .78]) (Fig. 3f), but remains strongly correlated. Against individual raters, the model reached $r = .69$ (95% CI [.64, .73]) with the first rater (Fig. 3g) and $r = .72$ (95% CI [.67, .76]) with the second rater (Fig. 3h). All correlations were significant at $p < .001$. While correlations for external details were slightly weaker than those in the training holdout data, the model's scores remained consistently aligned with human ratings.

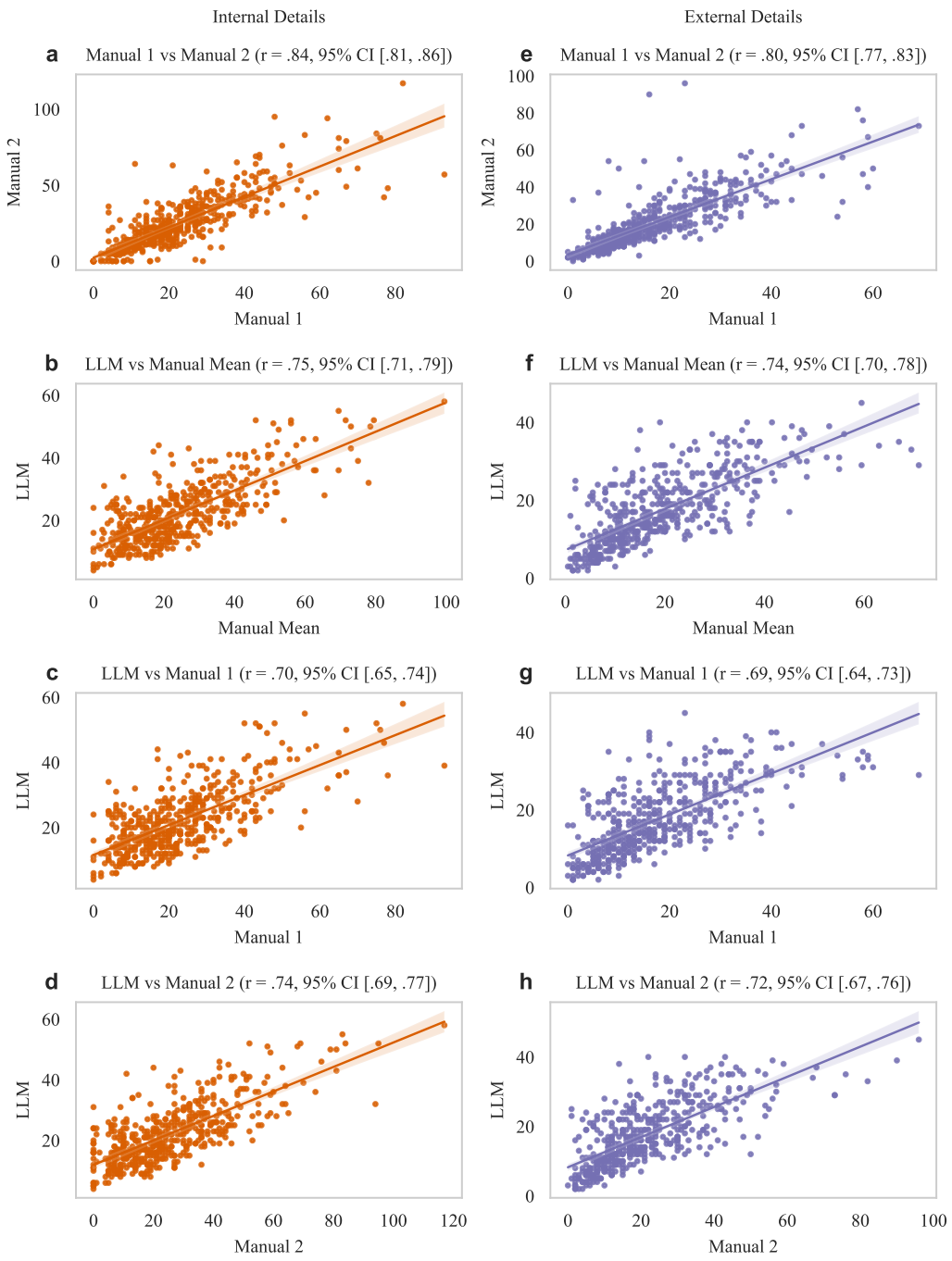

Figure 3. Model alignment with human raters on internal and external detail scoring in the past autobiographical event thinking task of Dataset C. For internal details, the model's alignment with the average human score (b), the first individual scorer (c), and the second individual scorer (d) shows strong alignment, though slightly lower than the high benchmark set by the alignment between the two human raters (a). For external details, the model's alignment with the average human score (f) and each individual scorer (g-h) similarly demonstrates robust alignment, though slightly lower than the alignment observed between human raters (e). Given that Dataset C is a novel dataset beyond the training data, this loss in prediction accuracy is to be expected due to potential differences in samples, context, or time. All correlations are significant at $p < .001$.

Dataset C: Future autobiographical event thinking

In the future autobiographical event thinking task from Dataset C, the model continued to demonstrate strong alignment with human raters. For internal details, the model achieved a correlation of r = .75 (95% CI [.70, .78]) with the human mean, r = .69 (95% CI [.64, .73]) with the first scorer, and r = .72 (95% CI [.67, .76]) with the second scorer, while the two human raters showed a correlation of r = .78 (95% CI [.75, .82]) (Fig. 2a). For external details, the model's scores showed a correlation of r = .71 (95% CI [.66, .75]) with the human mean, r = .68 (95% CI [.63, .72]) with the first scorer, and r = .68 (95% CI [.63, .73]) with the second rater, while the correlation between the human raters was r = .83 (95% CI [.80, .86]) (Fig. 2b). All correlations were significant at $p < .001$. While these correlations are somewhat lower compared to prior tasks, they nonetheless reflect strong model performance in scoring narratives related to future-oriented thinking, further supporting the model's ability to generalize across tasks. A detailed visualization of the correlations using scatter plots, similar to those presented for the past autobiographical event thinking task in Fig. 3, is provided in Supplementary Fig. 3.

Dataset C: Autobiographical Interview

In the Autobiographical Interview from Dataset C, the model's performance for internal details remained strong, achieving correlations of r = .74 (95% CI [.70, .78]) with the human mean, r = .71 (95% CI [.66, .75]) with the first scorer, and r = .70 (95% CI [.65, .74]) with the second scorer, while human raters showed a correlation of r = .81 (95% CI [.78, .84]) (Fig. 2a). However, for external details, the model's performance was only moderate, with correlations of r = .61 (95% CI [.55, .66]) with the human mean, r = .55 (95% CI [.48, .61]) with the first scorer, and r = .59 (95% CI [.52, .64]) with the second scorer, compared to a human rater correlation of r = .76 (95% CI [.72, .80]) (Fig. 2b). All correlations were significant at $p < .001$. While this represents the model's lowest alignment for external details across all datasets, the

model maintained strong performance on internal details. A detailed visualization of the correlations using scatter plots, similar to those presented for the past autobiographical event thinking task in Fig. 3, is provided in Supplementary Fig. 4.

Factors influencing model-human score discrepancies

For Dataset C, we explored factors that might account for differences between the model's scores and the average manual scores. We examined whether discrepancies were associated with participant variables, such as age and education level, total word count of the narrative, or with the degree of disagreement between manual raters, quantified as the absolute difference between their scores. The regression model had low explanatory power (R^2 = .07), indicating that the included factors account for only a small portion of the variance in the model-human discrepancies. Individual correlations also showed small effect sizes, with all correlation coefficients below .2. When examining participant variables, education level did not show a significant effect on model-human score differences (r = .01, p = .653), while age had a small positive correlation with discrepancies (r = .11, p < .001). Total word count showed a small negative correlation with discrepancies (r = -.12, p < .001), which does not suggest that the model relies on narrative length as a simplified proxy for predicting the number of details. Interestingly, one of the strongest correlations was with the level of divergence between human raters (r = .15, p < .001). This suggests that the model tends to deviate more from the human average when the two human raters themselves exhibit less alignment. This pattern may indicate that the model is more likely to diverge from human scores in cases where human judgments are inconsistent, potentially reflecting the model's sensitivity to ambiguity in the scoring.

Evaluation of scoring robustness with preprocessed narratives

To assess whether the model primarily focuses on relevant narrative content or incorporates irrelevant elements and noise from unprocessed transcripts, we conducted an exploratory analysis to evaluate the impact of preprocessing on scoring outcomes. We applied preprocessing steps, including the removal of interviewer speech and fillers, the replacement of placeholders for sensitive information, and punctuation correction, to all three tasks from Dataset C. This preprocessing was entirely automated and rule-based, using Python packages, meaning it did not necessarily achieve the same processing level as a human and instead served as an exploratory additional robustness check. We ran the automated scoring on these processed transcripts and compared the results to those from the original unprocessed narratives. Correlations between automated scores for unprocessed and processed transcripts were very high across all three tasks, $r = .89$ for the past event thinking task, $r = .90$ for the future event thinking task, and $r = .88$ for the Autobiographical Interview for internal details, and $r = .86$, .87, and .84, respectively, for external details. All correlations were significant at $p < .001$. Minimal impact from preprocessing suggests that the model reliably focuses on relevant narrative content to identify internal and external details while disregarding noise.

Age group differences

Research on autobiographical memory consistently shows that cognitively healthy older adults recall fewer internal details and more external details than young adults, a pattern widely documented in studies on memory specificity and cognitive aging (Levine et al., 2002; Simpson et al., 2023). To further validate our model, we aimed to replicate this finding using automated scoring in comparison to manual scorers. Beyond correlation measures, capturing these age group differences provides a complementary test of the model's robustness and practical utility. By assessing the model's ability to identify age-related differences in internal and external

details using the independent Dataset C, which includes both past and future thinking, we can evaluate its effectiveness in detecting established patterns in memory research.

Dataset C: Past autobiographical event thinking

For the past autobiographical event thinking task in Dataset C, we analyzed the interaction effects across all four scorings with detail type as a within-subject factor and age group as a between-subject factor, using a two-way ANOVA. Significant interaction effects were observed for both the first manual scorer ($F(1,97) = 35.44$, $p < .001$, partial $\eta^2 = .27$) and the second manual scorer ($F(1,97) = 31.51$, $p < .001$, partial $\eta^2 = .25$), their average ($F(1,97) = 36.05$, $p < .001$, partial $\eta^2 = .27$), and the automated model ($F(1,97) = 33.77$, $p < .001$, partial $\eta^2 = .26$). Given these consistently significant interaction effects across scoring methods, we proceeded to examine the simple effects in more depth.

For internal details, all scorers reflected lower scores for older adults compared to young adults. The model produced a mean score of 25.9 (*SD* = 9.3) for young adults and 21.4 (*SD* = 9.6) for older adults, with a significant mean difference of 4.5 details ($t = 4.00$, $p < .001$, $d = 0.47$). The first manual scorer recorded means of 32.1 (*SD* = 13.1) for young adults and 21.0 (*SD* = 14.4) for older adults, yielding a significant difference of 11.1 details ($t = 6.93$, $p < .001$, $d = 0.78$). The second manual scorer showed similar results, with young adults scoring a mean of 36.3 (*SD* = 14.5) and older adults 22.9 (*SD* = 17.5), resulting in a significant difference of 13.4 details ($t = 7.40$, $p < .001$, $d = 0.79$). The manual scorer mean reflected this age difference as well, with young adults at 34.2 (SD = 12.9) and older adults at 22.0 (SD = 15.3), producing a mean difference of 12.2 details (t = 7.62, p < .001, d = 0.82) (Fig. 4a). These results indicate that both the model and manual raters captured the expected reduction in internal details for older adults.

For external details, older adults scored higher across all scorers, aligning with expected age-related trends. The model yielded a mean score of 13.9 (*SD* = 8.8) for young adults and

18.2 (*SD* = 8.8) for older adults, with a significant mean difference of 4.3 details (t = -4.04, p < .001, d = -0.49). The first manual scorer recorded means of 13.5 (*SD* = 11.7) for young adults and 18.3 (*SD* = 11.5) for older adults, showing a significant difference of 4.8 details (t = -3.42, p < .001, d = -0.41). The second manual scorer showed a mean of 15.5 (*SD* = 11.4) for young adults and 22.5 (*SD* = 15.2) for older adults, with a difference of 7.0 details (t = -4.81, p < .001, d = -0.48). The manual scorer mean was also significantly different, with young adults at 14.5 (*SD* = 11.1) and older adults at 20.4 (*SD* = 12.7), resulting in a mean difference of 5.9 details (t = -4.30, p < .001, d = -0.48) (Fig. 4b). Together, these findings demonstrate that both the model and manual raters successfully replicated age-related differences in both internal and external details, underscoring the model's reliability in detecting established patterns observed in cognitive aging research. Interestingly, the effect size of the automated scoring appears smaller than that of manual scorers for internal details but is comparable, or even marginally higher, for external details.

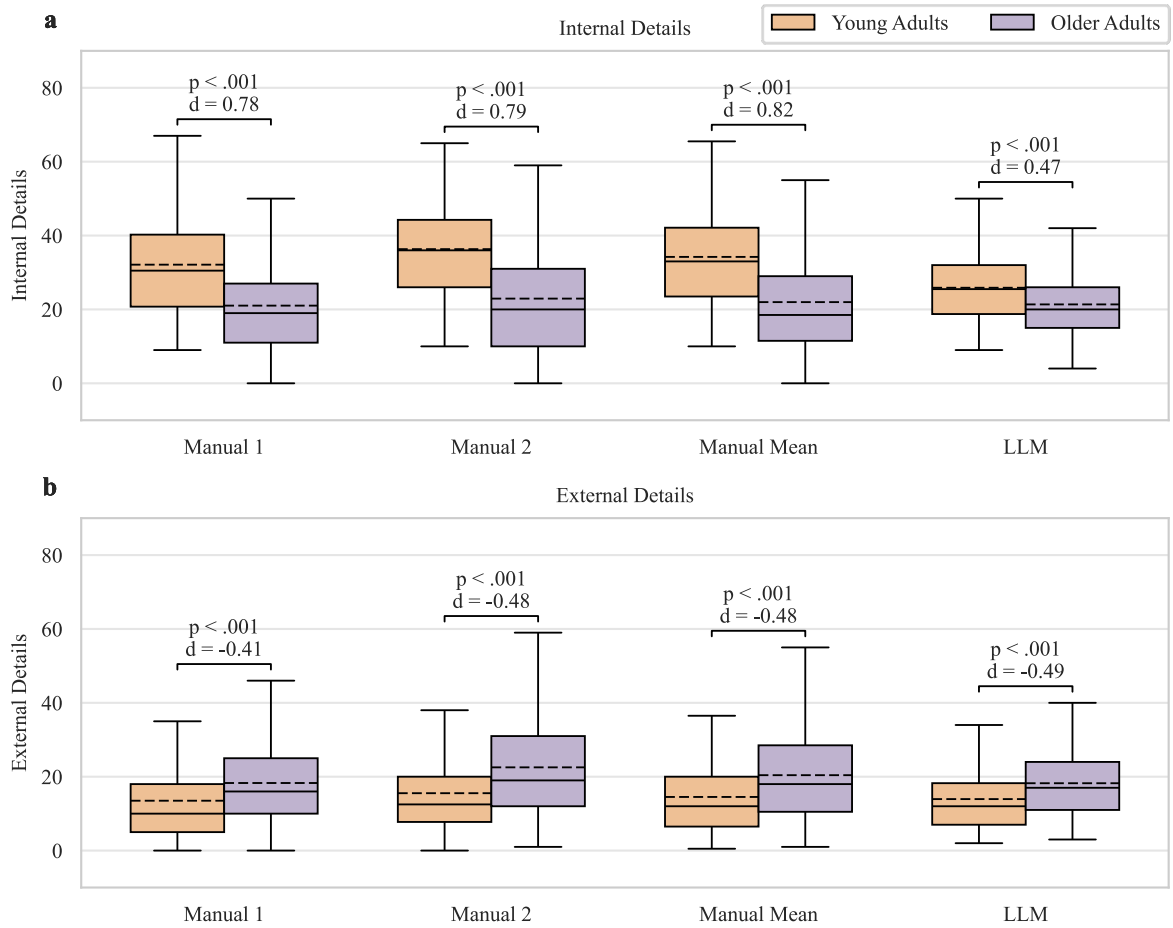

Figure 4. Age-related differences in internal and external detail scoring for the past autobiographical event thinking task in Dataset C. (a) Internal details scored by the first manual scorer, second manual scorer, manual mean, and automated model, showing significantly lower scores for older adults compared to young adults across all methods. (b) External details scored by the first manual scorer, second manual scorer, manual mean, and automated model, showing significantly higher scores for older adults compared to young adults across all methods. The solid horizontal line within each box represents the median, while the dashed horizontal line indicates the mean. Statistical significance is based on independent sample *t*-tests, with effect sizes reported as Cohen's *d*.

Dataset C: Future autobiographical event thinking

For the future thinking task in Dataset C, significant interactions were observed for both internal and external details across scorers, including the first manual scorer ($F(1,97) = 14.62$, $p < .001$, partial $\eta^2 = 0.13$), the second manual scorer ($F(1,97) = 17.14$, $p < .001$, partial $\eta^2 =$

0.15), their mean ($F(1,97) = 17.87$, $p < .001$, partial $\eta^2 = 0.16$), and our model ($F(1,97) = 15.66$, $p < .001$, partial $\eta^2 = 0.14$).

For internal details, young adults consistently scored higher than older adults across all scorers. Our model produced a mean score of 21.7 (*SD* = 8.8) for young adults and 17.3 (*SD* = 8.2) for older adults, with a significant mean difference of 4.4 details ($t = 4.31$, $p < .001$, $d = 0.54$). The first manual scorer recorded means of 23.5 (*SD* = 10.7) for young adults and 14.6 (*SD* = 10.9) for older adults, yielding a significant difference of 8.9 details ($t = 6.95$, $p < .001$, $d = 0.82$). The second manual scorer showed similar results, with young adults scoring a mean of 26.7 (*SD* = 13.3) and older adults 15.7 (*SD* = 13.0), resulting in a significant difference of 11.0 details ($t = 7.01$, $p < .001$, $d = 0.85$). The manual scorer mean also reflected this age difference, with young adults at 25.1 (*SD* = 11.2) and older adults at 15.1 (*SD* = 11.2), producing a mean difference of 10.0 details ($t = 7.48$, $p < .001$, $d = 0.89$) (Table 1). These results confirm that both the model and manual raters successfully captured the expected reduction in internal details for older adults in the future thinking task.

For external details, young adults consistently scored lower than older adults across all scorers, but only the model detected this difference significantly at a robust level. The model's mean score for young adults was 13.0 (*SD* = 7.5) compared to 15.5 (*SD* = 8.1) for older adults, yielding a significant mean difference of 2.5 details ($t = -2.81$, $p = .006$, $d = -0.32$). The first manual scorer recorded means of 13.3 (*SD* = 10.6) for young adults and 15.0 (*SD* = 10.8) for older adults, with a non-significant mean difference of 1.7 details ($t = -1.34$, $p = .183$, $d = -0.16$). The second manual scorer showed a mean of 14.4 (*SD* = 9.9) for young adults and 17.1 (*SD* = 12.7) for older adults, resulting in a significant mean difference of 2.7 details ($t = -2.13$, $p = .035$, $d = -0.22$). For the manual mean, young adults scored 13.8 (*SD* = 9.9) compared to 16.0 (*SD* = 11.3) for older adults, resulting in a difference of 2.2 details, though this did not reach significance ($t = -1.80$, $p = .075$, $d = -0.20$) (Table 1). These findings highlight the

model's sensitivity in detecting the established age-related increase in external details, even when manual raters did not consistently observe significant differences. Together, these results support the model's generalizability to future thinking as a different narrative context and reinforce its reliability in capturing known patterns in cognitive aging research. As with the past autobiographical event thinking task, the effect size for automated scoring is smaller than that of manual scorers for internal details. However, for external details, the effect size of automated scoring is notably larger.

A detailed visualization of the age group differences using box plots, similar to those shown for the past autobiographical event thinking task in Fig. 4, is provided in Supplementary Fig. 5.

Dataset C: Autobiographical Interview

For the Autobiographical Interview in Dataset C, significant interactions were observed for both internal and external details across scorers, including the first manual scorer ($F(1,97) = 11.47$, $p = .001$, partial $\eta^2 = 0.11$), the second manual scorer ($F(1,97) = 22.74$, $p < .001$, partial $\eta^2 = 0.19$), their mean ($F(1,97) = 20.12$, $p < .001$, partial $\eta^2 = 0.17$), and our model ($F(1,97) = 16.13$, $p < .001$, partial $\eta^2 = 0.14$).

For internal details, young adults consistently scored higher than older adults across all scorers. Our model produced a mean score of 31.0 (*SD* = 13.8) for young adults and 27.0 (*SD* = 11.4) for older adults, with a significant mean difference of 4.0 details ($t = 2.54$, $p = .013$, $d = 0.34$). The first manual scorer recorded means of 38.9 (*SD* = 21.6) for young adults and 30.6 (*SD* = 17.9) for older adults, yielding a significant difference of 8.3 details ($t = 3.35$, $p = .001$, $d = 0.45$). The second manual scorer showed similar results, with young adults scoring a mean of 40.6 (*SD* = 20.6) and older adults 30.6 (*SD* = 17.1), resulting in a significant difference of 10.0 details ($t = 4.28$, $p < .001$, $d = 0.57$). The manual scorer mean also reflected this age difference, with young adults at 39.8 (*SD* = 20.6) and older adults at 30.6 (*SD* = 16.5),

producing a mean difference of 9.2 details ($t = 3.92$, $p < .001$, $d = 0.53$) (Table 1). These results confirm that both the model and manual raters successfully captured the expected reduction in internal details for older adults in the autobiographical task.

For external details, young adults consistently scored lower than older adults across all scorers. The model's mean score for young adults was 16.1 (*SD* = 7.0) compared to 19.7 (*SD* = 8.0) for older adults, yielding a significant mean difference of 3.5 details ($t = -4.17$, $p < .001$, $d = -0.45$). The first manual scorer recorded means of 17.4 (*SD* = 11.8) for young adults and 20.8 (*SD* = 13.8) for older adults, with a significant mean difference of 3.4 details ($t = -2.37$, $p = .019$, $d = -0.25$). The second manual scorer showed a similar trend, with young adults scoring a mean of 20.3 (*SD* = 12.9) and older adults 27.2 (*SD* = 17.1), resulting in a significant mean difference of 6.9 details ($t = -4.26$, $p < .001$, $d = -0.42$). For the manual mean, young adults scored 18.9 (*SD* = 11.8) compared to 24.0 (*SD* = 14.5) for older adults, with a significant difference of 5.1 details ($t = -3.54$, $p < .001$, $d = -0.37$) (Table 1). These findings demonstrate that both the model and manual raters consistently detected the established age-related increase in external details, reinforcing the model's reliability in identifying known patterns in cognitive aging research across autobiographical narratives. Interestingly, while the same pattern observed in the previous tasks remains present in this task, namely lower effect sizes for the automated scoring on internal details and higher effect sizes on external details, the differences appear smaller for both in the present task.

A detailed visualization of the age group differences using box plots, similar to those shown for the past autobiographical event thinking task in Fig. 4, is provided in Supplementary Fig. 6.

Table 1. Age-related differences in internal and external details across past event thinking, future event thinking, and Autobiographical Interview tasks in Dataset C. The table presents the mean, standard deviation (*SD*), *p* values derived from independent samples *t*-tests, and effect sizes (Cohen's *d*) for internal and external details scored by the first manual scorer, second manual scorer, manual mean, and automated model (LLM) for both young and older adults. Significant differences between age groups are indicated across tasks, with older adults showing lower internal detail scores and higher external detail scores compared to young adults. This trend is consistent across scoring methods and aligns with established patterns in cognitive aging research.

	Internal Details						External Details					
	Young Adults		Older Adults				Young Adults		Older Adults			
	Mean	SD	Mean	SD	p	d	Mean	SD	Mean	SD	p	d
Past Event Thinking												
Manual 1	32.1	13.1	21.0	14.4	< .001	0.78	13.5	11.7	18.3	11.5	< .001	-0.41
Manual 2	36.3	14.5	22.9	17.5	< .001	0.79	15.5	11.4	22.5	15.2	< .001	-0.48
Manual Mean	34.2	12.9	22.0	15.3	< .001	0.82	14.5	11.1	20.4	12.7	< .001	-0.48
LLM	25.9	9.3	21.4	9.6	< .001	0.47	13.9	8.8	18.2	8.8	< .001	-0.49
Future Event Thinking												
Manual 1	23.5	10.7	14.6	10.9	< .001	0.82	13.3	10.6	15.0	10.8	.183	-0.16
Manual 2	26.7	13.3	15.7	13.0	< .001	0.85	14.4	9.9	17.1	12.7	.035	-0.22
Manual Mean	25.1	11.2	15.1	11.2	< .001	0.89	13.8	9.9	16.0	11.3	.075	-0.20
LLM	21.7	8.8	17.3	8.2	< .001	0.54	13.0	7.5	15.5	8.1	.006	-0.32
Autobiographical Interview												
Manual 1	38.9	21.6	30.6	17.9	.001	0.45	17.4	11.8	20.8	13.8	.019	-0.25
Manual 2	40.6	20.6	30.6	17.1	< .001	0.57	20.3	12.9	27.2	17.1	< .001	-0.42
Manual Mean	39.8	20.6	30.6	16.5	< .001	0.53	18.9	11.8	24.0	14.5	< .001	-0.37
LLM	31.0	13.8	27.0	11.4	.013	0.34	16.1	7.0	19.7	8.0	< .001	-0.45

Discussion

In this study, we introduced a novel automated scoring model for the Autobiographical Interview, leveraging advancements in large language models to process narratives without any data preprocessing. Our primary objectives were to assess whether the model could accurately identify internal and external details in autobiographical memory tasks, achieving a level of alignment with human scorers comparable to that observed between two human scorers. Additionally, we aimed to show that the model can replicate typical research findings related to age differences in narrative detail generation, where older adults produce fewer internal details and more external details compared to young adults (Levine et al., 2002; Simpson et al., 2023). We also examined the model's capacity to generalize to future thinking tasks, testing its

applicability beyond autobiographical memory and evaluating whether it could capture similar age-related patterns in a different context (Addis et al., 2008; Schacter & Addis, 2007). The results demonstrated that our model not only aligns closely with manual scoring but also generalizes effectively across different datasets and tasks, including future thinking. This held true for test holdout sets from the original training studies, as well as for an entirely novel dataset, underscoring the model's generalizability. Furthermore, our study replicates age-related distinctions in detail types, emphasizing its potential to transform narrative analysis in cognitive research by automating processes that were previously limited by time-consuming manual processes.

The automated scoring model exhibited strong correlations with human raters across multiple datasets, achieving alignment levels on internal and external detail scoring comparable to the alignment observed between human scorers. Notably, in Dataset A (Hernandez et al., 2024), the model's alignment with human ratings for internal details was comparable to, and in some cases exceeded, the consistency observed between human raters. Although the model's performance varied slightly across datasets—particularly in Dataset C, where alignment for external details was modestly lower—its efficiency and generalizability represent significant contributions. The ability to process raw, unprocessed transcripts, including interviewer speech, fillers, and timestamps, without the need for meticulous preprocessing, greatly enhances the practicality of large-scale research. This efficiency reduces the resource burden associated with manual scoring and enables the inclusion of larger and more diverse participant samples.

A critical test of the model's robustness was its ability to replicate well-established age-related differences in narrative detail generation. Consistent with previous research (Levine et al., 2002; Simpson et al., 2023), our automated scoring showed that cognitively normal older adults produced fewer internal details and more external details compared to young adults

across all tasks in Dataset C. The model detected significant reductions in internal details for older adults in a past autobiographical event thinking task, a future autobiographical event thinking task, and the traditional Autobiographical Interview. Additionally, the model identified significant increases in external details for older adults across these tasks, including in the future thinking task, where manual scorers did not consistently observe this difference. This enhanced sensitivity in capturing age-related patterns, even where human raters showed inconsistency, confirms the validity of the automated scoring method and highlights its potential as a reliable tool for cognitive aging research, offering insights comparable to manual scoring.

Compared to earlier methods, such as the sentence-level scoring approach by van Genugten & Schacter (2024), our model offers several key advancements. Firstly, it eliminates the need for preprocessing, handling raw transcripts that include interviewer prompts, filler words, and transcription artifacts. This reduces the manual effort required and minimizes potential errors introduced during preprocessing. Secondly, our model operates at the narrative level rather than the sentence or word level, capturing the relationships between words and sentences within the entire narrative. This holistic approach allows the model to consider the broader context, potentially enhancing its ability to distinguish between internal and external details. Similar to van Genugten & Schacter (2024), the model was trained on datasets that varied in their methods. For example, these included both past and future thinking tasks and data collected both in the lab and at home via videoconference interviews. This variability likely enhances the model's robustness to generalize to a diverse set of tasks. Lastly, by utilizing advanced techniques like LoRA and quantization, we optimized the model for computational efficiency without sacrificing performance. This enables the application of the model on standard hardware, making it accessible for a wide range of research settings. As an example, the fully automated scoring of one narrative (i.e., one memory or future event) on a single

NVIDIA RTX A6000 GPU with 48 GB of memory, as used in our training, takes approximately five seconds, compared to the five to ten minutes required by a manual scorer (van Genugten & Schacter, 2024).

To provide access to the model, we created a Jupyter Notebook with all necessary code and clear instructions. Researchers can run this notebook locally or upload it to platforms like Google Colab, requiring only a spreadsheet with narratives in CSV format to start automatic scoring. The notebook, along with setup instructions and a sample data file, will be made available upon publication of the manuscript and is also available beforehand upon request to the authors.

Limitations and future directions

Despite the model's strong performance, several limitations highlight areas for further investigation. First, the model’s alignment with human raters varied across datasets and tasks, with lower alignment for external details in Dataset C. This variability could be influenced by differences in narrative styles, participant demographics, or the inherent subjectivity in manual scoring. Another possible factor could be the impact of preprocessing, as it can shape the input data and potentially influence outcomes. While our exploratory analysis on Dataset C suggested that preprocessing—such as removing interviewer speech and fillers, replacing placeholders, and correcting punctuation—had minimal impact on scoring consistency, these findings require validation on other independent datasets. Moreover, other factors that could influence scoring outcomes remain unexamined. Future research could investigate ways to enhance consistency across diverse datasets, possibly by adapting the model to different narrative styles and demographic factors to increase generalizability.

Another limitation is that the model currently supports only English narratives, restricting its use in multilingual or non-English-speaking populations. Researchers might address this by using advanced translation tools like DeepL Translate (n.d.) to convert narratives into English,

though the impact of translation on narrative detail and scoring accuracy remains to be validated. Future studies should assess effectiveness of translation tools in preserving scoring reliability for non-English narratives. Alternatively, fine-tuning the model for other languages could eliminate reliance on translation tools. This would involve training on narrative datasets in the target language and leveraging multilingual pre-trained models like mBERT. Such efforts would enhance the model's cultural and linguistic relevance, enabling its application to a broader, global audience.

Moreover, exploring finer-grained detail categorization presents an opportunity to deepen narrative analysis. According to Levine et al. (2002), internal details in autobiographical narratives can be subdivided further, with episodic details encompassing subcategories such as event, place, time, perception, and emotion/thought elements. These subcategories provide a more nuanced understanding of narrative content by distinguishing specific types of descriptive information within a story. Future work could explore scoring these detail subtypes to enhance the model’s capability for in-depth narrative analysis.

Finally, a limitation in validating the model with age-related findings in this study is the reliance on an imbalanced sample in Dataset C, which had a lower proportion of young adults. This imbalance may impact the generalizability of age-related findings and could limit the precision of our conclusions. Although the model successfully replicated typical age-related patterns, future studies should ensure balanced age distributions to strengthen the validity of these findings and further clarify the model’s ability to detect subtle age-related differences in narrative details.

Conclusion

This study advances automated scoring in autobiographical memory research by introducing a model that processes raw narratives without preprocessing, aligns closely with human scoring, and generalizes across tasks and datasets. The model's efficiency and scalability have the

potential to transform narrative analysis in cognitive psychology, enabling larger and more diverse studies that were previously constrained by resource limitations. By replicating established age-related differences and aligning with cognitive aging research, the model contributes valuable insights to the understanding of memory processes. Its potential applications extend to early detection of cognitive decline and broader narrative analysis tasks. Continued innovation in leveraging large language models like ours holds promise for enhancing psychological assessments and increasing research efficiency.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.17605/OSF.IO/QRZ62.

Open Practices Statement The automated scoring code, along with instructions for applying the trained model, will be made available upon publication of the manuscript and is also available beforehand upon request to the authors. The repository will also include scripts for fine-tuning the language model, conducting statistical analyses, and generating the figures presented in this paper. Due to concerns about participant privacy, the narratives used in this study will not be made publicly available. Dataset A and Dataset B were made publicly available with their original publications. Dataset C is an ongoing project. The data will not be made available until the project is completed. The analyses reported in this paper were not preregistered.

Authors' contributions JK designed the study, fine-tuned the large language model, conducted the statistical analyses, visualized the data, and wrote the original manuscript. DEC advised on the methodology and reviewed and edited the manuscript. ANG coordinated data collection and manual scoring and reviewed and edited the manuscript. MRM advised on the methodology and reviewed and edited the manuscript. JRAH advised on the methodology, secured funding, and reviewed and edited the manuscript. MDG designed the study, supervised all aspects of the work, secured funding, and reviewed and edited the manuscript.

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Declarations

Competing interests The authors declare no competing interests.

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